IIIT Hyderabad **Scribed By: Anupam Dwivedi (2025201032)**
- **Name 2 (Roll No. 2)**
Optimization Methods (CS1.404) **Lecture #Num**
Instructor: Dr. Naresh Manwani 2nd **Apr 2026**

Outline. This lecture covers the foundational theory of constrained optimization. We study feasible and descent directions, characterize local minima via the interaction of these direction sets, introduce active constraints, and build toward a matrix-form necessary condition. We conclude with Farkas' Lemma and its corollary, which provides the algebraic machinery underlying the KKT conditions.

1 Feasible and Descent Directions in Constrained Optimization

We consider the general constrained optimization problem:

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad g_i(x) \leq 0, \quad i = 1, \ldots, m, \tag{1}$$

where $f, g_i : \mathbb{R}^n \to \mathbb{R}$ are continuously differentiable. The *feasible set* is

$$\mathcal{X} = \{x \in \mathbb{R}^n : g_i(x) \leq 0,\ i = 1, \ldots, m\}.$$

Definition 1 (Feasible Direction)**.** A nonzero vector $d \in \mathbb{R}^n$ is called a **feasible direction** at a point $x \in \mathcal{X}$ if there exists $\bar{\alpha} > 0$ such that

$$x + \alpha d \in \mathcal{X} \quad \text{for all } \alpha \in (0, \bar{\alpha}).$$

Intuitively, d points *into* the feasible set from x; moving a small step along d keeps us feasible.

Definition 2 (Descent Direction)**.** A nonzero vector $d \in \mathbb{R}^n$ is called a **descent direction** for f at x if there exists $\bar{\alpha} > 0$ such that

$$f(x + \alpha d) < f(x) \quad \text{for all } \alpha \in (0, \bar{\alpha}).$$

That is, moving along d from x strictly decreases the objective.

Analytic characterisation of descent directions. When f is differentiable, by a first-order Taylor expansion,

$$f(x + \alpha d) = f(x) + \alpha \nabla f(x)^\top d + o(\alpha).$$

For small $\alpha > 0$ the sign of $f(x + \alpha d) - f(x)$ is determined by $\nabla f(x)^\top d$. Hence:

$$d \text{ is a descent direction at } x \iff \nabla f(x)^\top d < 0.$$

Remark 1. We define the sets

$$\mathcal{F}(x) = \{d \neq 0 : d \text{ is a feasible direction at } x\},$$
$$\mathcal{D}(x) = \{d \neq 0 : \nabla f(x)^\top d < 0\}.$$

A point $x \in \mathcal{X}$ can be improved (moved to a strictly feasible and strictly better point) if and only if $\mathcal{D}(x) \cap \mathcal{F}(x) \neq \emptyset$.

2 Characterization of Local Minima in Constrained Optimization

Definition 3 (Local Minimum)**.** A point $x^* \in \mathcal{X}$ is a **local minimum** of (1) if there exists $\delta > 0$ such that

$$f(x^*) \leq f(x) \quad \text{for all } x \in \mathcal{X} \text{ with } \|x - x^*\| < \delta.$$

Theorem 1 (Necessary Condition for Local Minimum — Direction Form)**.** *If x^* is a local minimum of (1) and f is continuously differentiable, then there is **no** direction d that is simultaneously a feasible direction and a descent direction at x^*. Equivalently,*

$$\mathcal{D}(x^*) \cap \mathcal{F}(x^*) = \emptyset.$$

Proof. Suppose for contradiction that some d satisfies $\nabla f(x^*)^\top d < 0$ and $x^* + \alpha d \in \mathcal{X}$ for all small $\alpha > 0$. Then

$$f(x^* + \alpha d) = f(x^*) + \alpha \underbrace{\nabla f(x^*)^\top d}_{<0} + o(\alpha) < f(x^*)$$

for all sufficiently small $\alpha > 0$. This contradicts x^* being a local minimum, since $x^* + \alpha d$ is feasible and has a strictly smaller objective value. □ □

Geometric intuition. At a local minimum, the gradient $\nabla f(x^*)$ must form an obtuse (or right) angle with every feasible direction — there is no direction you can move along that simultaneously decreases f and remains feasible.

3 Active Constraints and the Descent-Feasibility Sets

3.1 Active Constraints

Definition 4 (Active Constraint)**.** The i-th constraint $g_i(x) \leq 0$ is said to be **active** (or *binding*) at $x \in \mathcal{X}$ if $g_i(x) = 0$. The **active set** at x is

$$\mathcal{A}(x) = \{i \in \{1, \ldots, m\} : g_i(x) = 0\}.$$

Constraints with $g_i(x) < 0$ are called **inactive** (or *slack*) at x.

Why do inactive constraints not matter locally? If $g_i(x) < 0$, then by continuity of g_i, we have $g_i(x + \alpha d) < 0$ for all sufficiently small α and any direction d. So inactive constraints impose no local restriction on feasible directions.

3.2 The Linearized Feasible Direction Set

To work analytically, we approximate feasible directions using first-order information of the active constraints.

Definition 5 (Linearized Feasible Cone)**.** The **linearized feasible cone** at $x \in \mathcal{X}$ is

$$\widetilde{\mathcal{F}}(x) = \{d \in \mathbb{R}^n : \nabla g_i(x)^\top d \leq 0 \ \ \forall\, i \in \mathcal{A}(x)\}.$$

This is the set of directions that (to first order) do not violate any active constraint.

Definition 6 (Negative Directional Derivative Set)**.** The set of descent directions for the objective is

$$\mathcal{D}(x) = \{d \in \mathbb{R}^n \setminus \{0\} : \nabla f(x)^\top d < 0\}.$$

This is the open half-space on the opposite side of the hyperplane $\{\nabla f(x)^\top d = 0\}$ from $\nabla f(x)$.

4 Relation Between $\mathcal{F}(x)$ and $\widetilde{\mathcal{F}}(x)$ — With Proof

In this section, we establish the relationship between the actual feasible directions $\mathcal{F}(x)$ and the linearized feasible directions. Note that while Definition 5 defines the closed linearized cone $\widetilde{\mathcal{F}}(x)$ (using ≤ 0), the following theorem specifically examines the set of directions strictly pointing into the interior of the constraints, which we denote as the *strict* linearized feasible cone:

$$\widetilde{\mathcal{F}}_{<}(x) = \{d \in \mathbb{R}^n : \nabla g_i(x)^\top d < 0 \ \ \forall\, i \in \mathcal{A}(x)\}.$$

Theorem 2. *At any point $x \in \mathcal{X}$, every strictly linearized feasible direction is a feasible direction. That is,*

$$\widetilde{\mathcal{F}}_{<}(x) \subseteq \mathcal{F}(x).$$

Proof. Suppose $\widetilde{\mathcal{F}}_{<}(x) \neq \emptyset$ and let $d \in \widetilde{\mathcal{F}}_{<}(x)$. By definition, for all active constraints $i \in \mathcal{A}(x)$, we have $g_i(x) = 0$ and the directional derivative $\nabla g_i(x)^\top d < 0$.
By the definition of the directional derivative, there exists a strictly positive scalar $\delta_i > 0$ such that moving along d locally decreases the value of g_i:

$$g_i(x + \alpha d) < g_i(x) = 0 \quad \text{for all } \alpha \in (0, \delta_i).$$

Let $\delta = \min_{i \in \mathcal{A}(x)} \delta_i$. Because $\mathcal{A}(x)$ is a finite set of indices, $\delta > 0$. Thus, for all $\alpha \in (0, \delta)$ and all active constraints $i \in \mathcal{A}(x)$, we have $g_i(x + \alpha d) < 0$.
Now, consider the inactive constraints. Let $\mathcal{I}(x) = \{1, 2, \ldots, m\} \setminus \mathcal{A}(x)$ denote the set of inactive constraints at x. For all $j \in \mathcal{I}(x)$, we have by definition $g_j(x) < 0$. Since g_j is continuous, the function value must remain strictly negative in a small neighborhood around x. That is, there exists some $\gamma > 0$ such that:

$$g_j(x + \alpha d) \leq 0 \quad \text{for all } \alpha \in (0, \gamma) \text{ and for all } j \in \mathcal{I}(x).$$

Finally, we reconcile the step sizes by defining $\beta = \min(\delta, \gamma)$. For any step size $\alpha \in (0, \beta)$, we simultaneously satisfy:

$$\begin{aligned} g_i(x + \alpha d) &< 0 \quad \forall\, i \in \mathcal{A}(x), \\ g_j(x + \alpha d) &\leq 0 \quad \forall\, j \in \mathcal{I}(x). \end{aligned}$$

Therefore, $g_k(x + \alpha d) \leq 0$ for all $k \in \{1, 2, \ldots, m\}$. This implies that $(x + \alpha d) \in \mathcal{X}$ for all $\alpha \in (0, \beta)$. By the standard definition of a feasible direction, this means $d \in \mathcal{F}(x)$. This concludes the proof that $\widetilde{\mathcal{F}}_{<}(x) \subseteq \mathcal{F}(x)$. $\square$

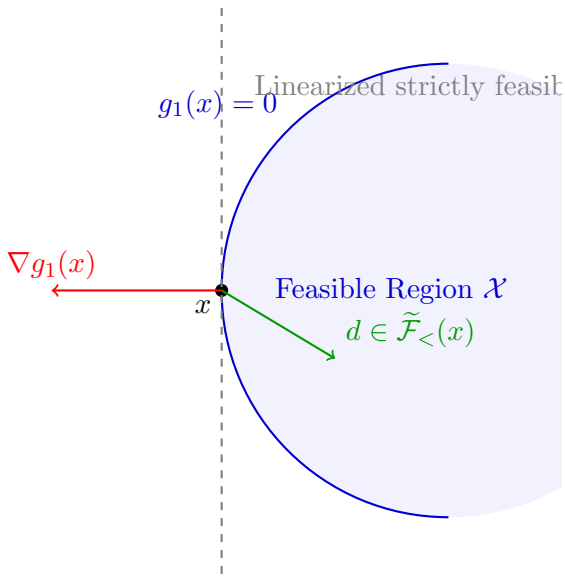

Figure 1: At point x, the gradient $\nabla g_1(x)$ points outwards (the direction of maximum objective increase). The condition $\nabla g_1(x)^\top d < 0$ defines the strict linearized feasible cone (the entire open half-space to the right of the dashed tangent line). As illustrated, any strictly linearized direction d inevitably falls inside the actual curved feasible set $\mathcal{X}$ after taking a sufficiently small step α.

Remark 2. A crucial consequence of $\widetilde{\mathcal{F}}_<(x) \subseteq \mathcal{F}(x)$ relates back to the optimality condition. If a point x^* is a local minimum, Theorem 1 states $\mathcal{D}(x^*) \cap \mathcal{F}(x^*) = \emptyset$. Since the strict linearized cone is a subset of the actual feasible cone, this naturally implies:

$$\widetilde{\mathcal{F}}_<(x^*) \cap \mathcal{D}(x^*) = \emptyset.$$

This forms the foundation for algebraically identifying local minima by inspecting gradient arrays.

5 Necessity versus Sufficiency of the Characterization

The condition $\widetilde{\mathcal{F}}_<(x^*) \cap \mathcal{D}(x^*) = \emptyset$ derived previously is a **necessary** condition for local optimality, but it is **not sufficient**. This means that finding a point x^* that satisfies this condition does not guarantee that x^* is actually a local minimum. To illustrate this and highlight the reliance on the functional form of the constraints, we consider two examples.

Example 1 (A Point Satisfying the Necessary Condition that is NOT a Local Minimum). Consider the optimization problem:

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & x_1^2 + x_2^2 \\ \text{subject to} \quad & h_1(x) = (x_1 + x_2 - 1)^3 \leq 0, \\ & x_1 \geq 0, \quad x_2 \geq 0. \end{aligned}$$

Geometrically, the feasible region is the solid triangle defined by $x_1 + x_2 \leq 1$ in the first quadrant. The objective represents the squared distance from the origin. Clearly, the true global minimum is at the origin $(0, 0)$.

Consider any point $x_A = (a, b)$ on the boundary line segment where $a + b = 1$ (with $a > 0, b > 0$). At this point, the active constraint is $h_1(x)$. Let us compute the gradient of h_1 at x_A:

$$\nabla h_1(x) = \begin{bmatrix} 3(x_1 + x_2 - 1)^2 \\ 3(x_1 + x_2 - 1)^2 \end{bmatrix} \implies \nabla h_1(x_A) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The strictly linearized feasible cone at x_A is defined by:

$$\widetilde{\mathcal{F}}_<(x_A) = \{d \in \mathbb{R}^2 : \nabla h_1(x_A)^\top d < 0\} = \{d \in \mathbb{R}^2 : \mathbf{0}^\top d < 0\} = \emptyset.$$

Since $\widetilde{\mathcal{F}}_<(x_A)$ is empty, its intersection with the descent cone $\mathcal{D}(x_A)$ is trivially empty:

$$\widetilde{\mathcal{F}}_<(x_A) \cap \mathcal{D}(x_A) = \emptyset.$$

Thus, x_A perfectly satisfies the necessary condition for a local minimum. Yet, we know x_A is emphatically *not* a local minimum (from x_A, we can move straight down inside the triangle to decrease the objective). The failure here is that $\nabla h_1(x_A) = 0$ provides no local first-order geometric information about the boundary.

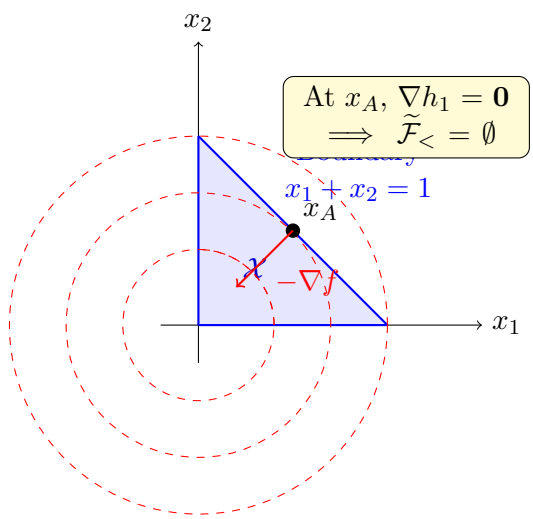

Figure 2: Visualizing Example 1: The point x_A satisfies the necessary optimality conditions because the active constraint's gradient vanishes. However, the objective can clearly decrease by moving towards the origin within $\mathcal{X}$, proving the condition is not sufficient.

Example 2 (A Point Failing the Necessary Condition Appropriately). Let us slightly modify the previous problem by replacing the cubic constraint with a linear one representing the *exact same geometric feasible set*:

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & x_1^2 + x_2^2 \\ \text{subject to} \quad & h_1(x) = (x_1 + x_2 - 1) \leq 0, \\ & x_1 \geq 0, \quad x_2 \geq 0. \end{aligned}$$

Consider the same point $x_A = (a, b)$ with $a + b = 1$. Now, the gradient of the active constraint is:

$$\nabla h_1(x_A) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The strictly linearized feasible cone is no longer empty:

$$\widetilde{\mathcal{F}}_<(x_A) = \left\{ d \in \mathbb{R}^2 : d_1 + d_2 < 0 \right\} \neq \emptyset.$$

We can compute the descent cone for the objective. The gradient of $f(x) = x_1^2 + x_2^2$ is $\nabla f(x_A) = [2a, 2b]^\top$. So the descent cone is:

$$\mathcal{D}(x_A) = \left\{ d \in \mathbb{R}^2 : 2ad_1 + 2bd_2 < 0 \right\}.$$

It is trivial to find a direction d that belongs to both sets (e.g., $d = [-a, -b]^\top$, which is moving straight towards the origin, yielding $-a - b = -1 < 0$ and $-2a^2 - 2b^2 < 0$). Thus,

$$\widetilde{\mathcal{F}}_<(x_A) \cap \mathcal{D}(x_A) \neq \emptyset.$$

This correctly flags that x_A is *not* a local minimum, showing how deeply the necessary condition relies on a non-degenerate algebraic representation (non-vanishing gradients) of the constraints.

5.1 The Limitation Handling Equality Constraints

The necessary condition $\widetilde{\mathcal{F}}_<(x^*) \cap \mathcal{D}(x^*) = \emptyset$ is immensely powerful for inequality constraints $h_j(x) \leq 0$. However, it completely breaks down when applied naively to **equality constraints**. Let us examine why this happens.

Example 3 (Equality Constraints Lead to Trivial Linearized Cones). Suppose we have an equality constraint $x_1 + x_2 - 1 = 0$. A standard technique to handle an equality constraint is to split it into two opposing inequality constraints:

$$\begin{aligned} h_1(x) &= (x_1 + x_2 - 1) \leq 0 \\ h_2(x) &= -(x_1 + x_2 - 1) \leq 0 \end{aligned}$$

If a point x_A is feasible (meaning it lies on the line $x_1 + x_2 - 1 = 0$), then both h_1 and h_2 are tightly active at 0. Computing the gradients yields:

$$\nabla h_1(x_A) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \nabla h_2(x_A) = \begin{bmatrix} -1 \\ -1 \end{bmatrix}.$$

To find the strictly linearized feasible cone, a direction d must simultaneously satisfy both strict inner product conditions:

$$\begin{aligned} \nabla h_1(x_A)^\top d < 0 &\implies d_1 + d_2 < 0 \\ \nabla h_2(x_A)^\top d < 0 &\implies -(d_1 + d_2) < 0 \implies d_1 + d_2 > 0 \end{aligned}$$

It is mathematically impossible for $d_1 + d_2$ to be strictly less than 0 and strictly greater than 0 at the same time. Therefore,

$$\widetilde{\mathcal{F}}_<(x_A) = \emptyset.$$

Because the set is empty, $\widetilde{\mathcal{F}}_<(x_A) \cap \mathcal{D}(x_A) = \emptyset$ will hold trivially for *any* point on the feasibility line, turning the condition into a tautology regardless of whether the point is a local minimum or not.

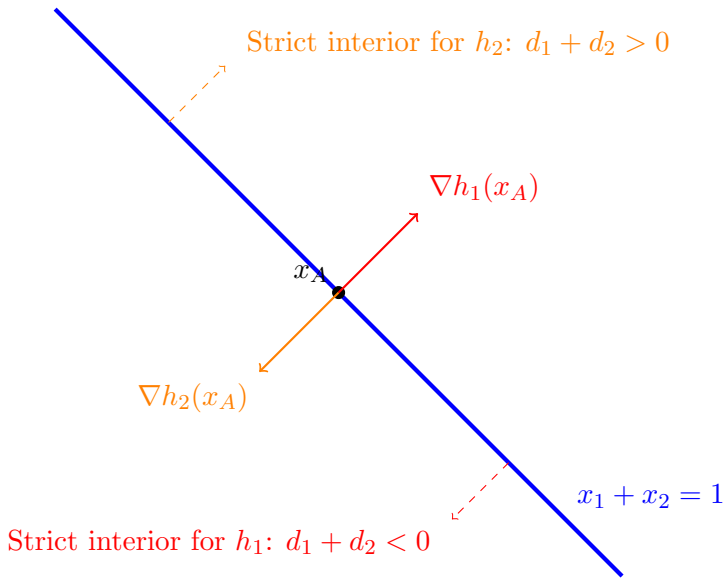

Figure 3: Visualizing Example 3: An equality constraint modelled as two inequalities. The strict feasible directions for h_1 point into the open half-space below the line, and those for h_2 point strictly above it. The intersection of these two open half-spaces is empty, trivially leading to $\widetilde{\mathcal{F}}_< = \emptyset$.

Important Limitation

The strict interior cone characterization $\widetilde{\mathcal{F}}_<(x^*) \cap \mathcal{D}(x^*) = \emptyset$ is **strictly applicable only for inequality constraints**. For systems containing equality constraints, we must rely on closed tangent cones or directly use the KKT gradient matrix stationarity conditions directly.

6 Matrix Representation of the Local Minimum Characterization

Let $x^* \in \mathcal{X}$ and let the active constraint index set be $\mathcal{A}(x^*) = \{i_1, i_2, \ldots, i_k\}$ for some $k \leq m$. Define the **active constraint gradient matrix**:

$$G = \begin{bmatrix} \nabla g_{i_1}(x^*)^\top \\ \nabla g_{i_2}(x^*)^\top \\ \vdots \\ \nabla g_{i_k}(x^*)^\top \end{bmatrix} \in \mathbb{R}^{k \times n}.$$

The linearized feasible cone and descent set become:

$$\widetilde{\mathcal{F}}(x^*) = \{d \in \mathbb{R}^n : Gd \leq 0\},$$
$$\mathcal{D}(x^*) = \{d \in \mathbb{R}^n : \nabla f(x^*)^\top d < 0\}.$$

Necessary condition in matrix form. The condition $\mathcal{D}(x^*) \cap \widetilde{\mathcal{F}}(x^*) = \emptyset$ states:

$$\boxed{\text{There is no } d \in \mathbb{R}^n \text{ such that } \nabla f(x^*)^\top d < 0 \text{ and } Gd \leq 0.}$$

By Farkas' Lemma (Theorem 1), this is equivalent to the existence of a vector $\lambda \in \mathbb{R}^k$, $\lambda \geq 0$, such that:

$$\nabla f(x^*) + G^\top \lambda = 0 \quad \Longleftrightarrow \quad \nabla f(x^*) = -\sum_{j=1}^{k} \lambda_{i_j} \nabla g_{i_j}(x^*).$$

Extending to all constraints with $\lambda_i = 0$ for $i \notin \mathcal{A}(x^*)$, we get the **KKT stationarity condition**:

$$\nabla f(x^*) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x^*) = 0, \qquad \lambda_i \geq 0, \quad \lambda_i g_i(x^*) = 0 \ \ \forall\, i.$$

The last condition $\lambda_i g_i(x^*) = 0$ (complementary slackness) follows because $\lambda_i = 0$ for all inactive constraints (by construction) and $g_i(x^*) = 0$ for all active ones.
Summary in compact matrix form.

Local Minimum Characterization (Matrix Form)

x^* is a local minimum of $\min f(x)$ s.t. $g_i(x) \leq 0$ only if there exist $\lambda \in \mathbb{R}^m$, $\lambda \geq 0$, such that:

$$\begin{aligned} \nabla f(x^*) + G^\top \lambda_{\mathcal{A}} &= 0 \quad \text{(Stationarity)}, \\ Gx^* &= 0 \quad \text{(Active constraint at } x^*\text{)}, \\ \lambda_{\mathcal{A}} &\geq 0 \quad \text{(Non-negativity of multipliers)}, \end{aligned}$$

where $G \in \mathbb{R}^{|\mathcal{A}(x^*)| \times n}$ is the active constraint gradient matrix and $\lambda_{\mathcal{A}}$ are the associated multipliers.

7 Farkas' Lemma and Its Translation to Optimality

The transition from the geometric condition $\widetilde{\mathcal{F}}_<(x^*) \cap \mathcal{D}(x^*) = \emptyset$ (derived in previous sections) to the algebraic KKT stationarity condition relies fundamentally on a theorem of alternatives known as Farkas' Lemma.

Theorem 3 (Farkas' Lemma)**.** *Let $A \in \mathbb{R}^{m \times n}$ be a given matrix, and $c \in \mathbb{R}^n$ a given vector. Then exactly one of the following two systems has a solution:*

1. *System 1: $Ax \leq 0$ and $c^\top x < 0$ for some $x \in \mathbb{R}^n$.*
2. *System 2: $A^\top y = c$ and $y \geq 0$ for some $y \in \mathbb{R}^m$.*

Geometrically, Farkas' lemma states that either the vector c lies within the convex cone generated by the rows of A (System 2), or there exists a hyperplane passing through the origin that strictly separates c from that cone (System 1).

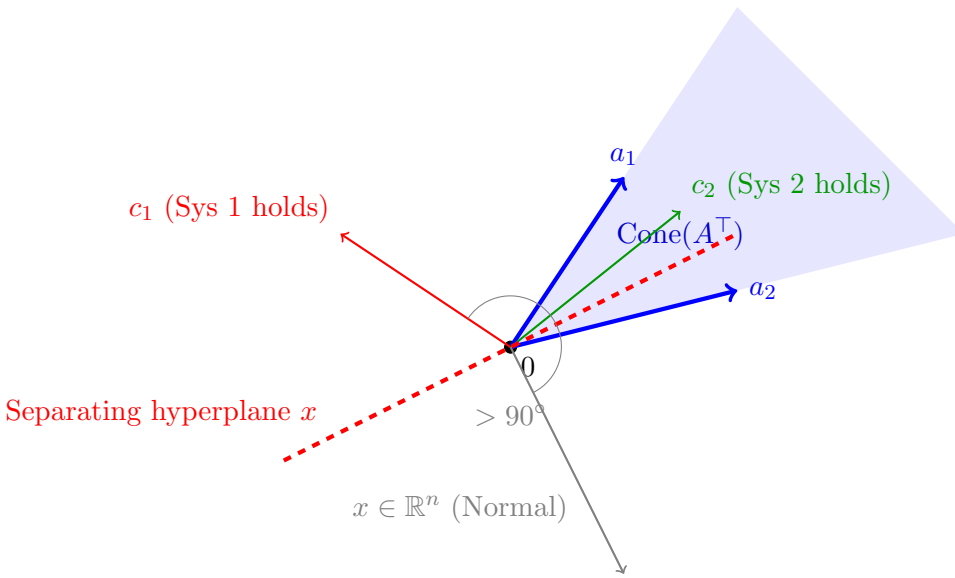

Figure 4: Visualizing Farkas' Lemma. For a vector c_2 situated inside the cone spanned by the rows of A (i.e. a_1, a_2), it can be represented as a non-negative linear combination $y_1 a_1 + y_2 a_2$ (System 2). Conversely, if c_1 is strictly outside the cone, we can find a strictly separating hyperplane. The normal vector to this hyperplane, x, acts as a direction making an obtuse angle with c_1 ($c_1^\top x < 0$) and obtuse/right angles with the cone generators ($a_i^\top x \leq 0 \implies Ax \leq 0$) (System 1).

Corollary 1 (Optimality via Farkas' Lemma)**.** *Let us apply Farkas' Lemma to our derived necessary condition $\widetilde{\mathcal{F}}_<(x^*) \cap \mathcal{D}(x^*) = \emptyset$.*
Recall this geometric condition means there exists no direction $d \in \mathbb{R}^n$ such that:

$$Gd \leq 0 \quad \textit{and} \quad \nabla f(x^*)^\top d < 0$$

where $G \in \mathbb{R}^{k \times n}$ is the matrix whose rows are the gradients of the active constraints $\nabla g_{i_j}(x^*)^\top$.
Identifying our variables with Farkas' Lemma:

- $x \longleftrightarrow d$ *(The search direction)*
- $A \longleftrightarrow G$ *(The active constraint normal matrix)*
- $c \longleftrightarrow \nabla f(x^*)$ *(The objective gradient)*

Since we established that System 1 has NO mathematical solution *at a strict local minimum point x^*, it logically necessitates that System 2* MUST have a solution.
Therefore, there exists a vector of non-negative multipliers $\lambda_\mathcal{A} \geq 0$ (corresponding to the active constraints) such that:

$$G^\top \lambda_\mathcal{A} = -\nabla f(x^*) \implies \nabla f(x^*) + G^\top \lambda_\mathcal{A} = \mathbf{0} \implies \nabla f(x^*) + \sum_{j=1}^{k} \lambda_{i_j} \nabla g_{i_j}(x^*) = \mathbf{0}.$$

This perfectly bridges the gap, translating the conceptual lack of geometric descent directions into the clean algebraic KKT stationarity equations that we solve in practice.

References